Kaushik's KAC3 Factor Risk Model

Methodology Notes -- Multi-Asset-Class Equity Factor Risk Modeling

Personal study notes on cross-sectional factor risk model construction

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Purpose: Working notes on the design of a modern multi-asset-class equity factor risk model -- written in my own words for personal study, to sharpen my understanding of portfolio risk analytics tooling used in institutional risk management, and to inform the companion Kaushik_KAC3_Factor_Model_Example.xlsx worked example.

Scope: Model overview and design philosophy, factor exposure construction, cross-sectional factor return estimation, term-structure volatility and correlation forecasting, the specific (idiosyncratic) risk model, the finite-sample adjustment for pure factor portfolios, and the bias/Q-statistic framework used to validate a risk model.

Sections	Style Factors	Prediction Horizons	Core Innovation
7 + Appendix	~14 (composite)	6 (daily to long-term)	Finite-sample adjustment

A note on sourcing: these notes paraphrase and re-derive, in my own words, the general methodology used by modern institutional multi-asset-class equity risk models. No vendor name, logo, or proprietary document text is reproduced here -- the content below reflects my own understanding of the underlying econometric techniques, which are widely discussed in the academic factor-model literature (Rosenberg 1974; Menchero and co-authors, various years).

1. Model Overview

1.1 Design Highlights

- **Term structure of risk.** Rather than forecasting a single volatility number, the model produces risk forecasts at six prediction horizons -- daily, weekly, monthly, quarterly, annual, and long-term -- using a mixed-frequency approach that blends high-frequency (daily) and low-frequency (weekly/monthly) return observations. Each horizon is calibrated with a different trade-off between responsiveness and stability, since a risk manager with a one-week horizon and an asset owner making decade-scale allocation decisions need very different properties from the same underlying model.
- **Daily re-estimation.** Factor exposures, factor returns, factor covariances, and specific risk forecasts are all re-estimated every trading day so the model reflects the latest available market information.
- **Non-binary industry and country betas.** Instead of assuming every stock in a sector has an identical (0/1) exposure to that sector, the model estimates stock-specific industry and country sensitivities via time-series regression. This materially improves explanatory power and reduces spurious correlation between specific returns and factor returns.
- **Expanded style-factor set.** The model I studied uses roughly 14 style factors, versus about 10 in an earlier generation -- most notably splitting a single "volatility" factor into two distinct, more powerful factors: market beta and residual volatility.
- **Modern industry classification.** A four-level industry hierarchy (sector to industry group to industry to sub-industry) is used to define industry factors, with the granularity chosen empirically to balance explanatory power against having too few names in a given industry bucket (which would make that factor's return statistically unreliable).
- **Inverse-variance regression weighting.** Traditional factor models weight the cross-sectional regression by the square root of market capitalization. Econometrically, the weighting that minimizes noise in the factor-return estimate is inverse residual variance, which the model uses instead -- reducing noise in factor returns and mitigating spurious factor/specific-return correlation.
- **Missing-factor imputation.** When a factor return is missing (e.g., due to a market holiday or a short history), the model imputes it from the other factors so the factor covariance matrix can still be estimated cleanly, rather than leaving a gap.

1.2 Local, Global, and Integrated Models

A model suite of this kind is typically built at three levels of aggregation. **Local models** use a detailed, market-specific factor structure for a single country. A **global model** spans all countries under one parsimonious set of global factors (a global market factor, global industries, global styles, plus country factors) -- convenient for a top-down read on, say, the aggregate momentum exposure of a worldwide portfolio, but it necessarily averages away country-specific nuance (momentum can behave very differently in Europe versus Asia in the same week, and a global momentum factor blurs that distinction). An **integrated model** tries to get the best of both: full global coverage while preserving the granular local factor structure within each market, by combining the individual local models with a global correlation structure spanning them.

1.3 A Short History of Factor Models

Modern Portfolio Theory began with Markowitz (1952), who framed portfolio construction as mean-variance optimization along an efficient frontier. Tobin (1958) showed that adding a risk-free asset collapses the efficient frontier into a straight line (the Capital Market Line), giving the two-fund separation result: every investor holds some mix of the risk-free asset and one common "super-efficient" portfolio. This is the intuition behind CAPM, which predicts that expected return is driven purely by market beta.

CAPM did not hold up well empirically. Basu (1977) found a return premium for high earnings-to-price (value) stocks; Banz (1981) documented a small-cap premium; Jegadeesh and Titman (1993) documented momentum; and Fama and French (1992) showed that market beta alone barely explains the cross-section of returns at all. These "anomalies" motivated active, factor-based investing: if the true drivers of return can be identified, a custom portfolio can be built with a better risk-adjusted return than a passive market holding.

The practical obstacle is that portfolio optimization needs a covariance matrix, and a raw sample covariance matrix estimated directly from stock returns is unreliable whenever the number of assets (N) is large relative to the number of time periods (T) -- a large-cap universe of a few thousand names estimated on a few years of daily data has N far exceeding T, which makes the sample covariance matrix rank-deficient and prone to "discovering" spurious risk-free combinations of risky assets. Rosenberg (1974) solved this by proposing a linear factor model, $r = Xf + u$, and estimating the much smaller KxK factor covariance matrix (K in the dozens) instead of the unreliable NxN asset covariance matrix directly -- a dimensionality reduction that is the foundation of every commercial factor risk model since.

Multi-asset-class integrated models reintroduce the original problem one level up: once equities, rates, credit, commodities, and FX are combined, the total factor count can run into the thousands, once again exceeding the available time-series history. Modern models handle this with a second layer of dimensional reduction (discussed in Section 5) when estimating the cross-asset-class factor correlation matrix.

1.4 Portfolio Risk and Return Decomposition

For a universe of N stocks and K factors, the core model is:

```
r = X f + u
```

where r is the Nx1 vector of stock excess returns, X is the NxK factor exposure matrix, f is the Kx1 vector of factor returns, and u is the Nx1 vector of specific (idiosyncratic) returns, assumed uncorrelated with the factors and mutually uncorrelated across stocks (with one notable exception discussed in Section 6.6 for linked securities of the same issuer). Given the KxK factor covariance matrix F = cov(f) and the NxN diagonal specific-variance matrix Delta = cov(u), the asset covariance matrix is:

```
Omega = X F X' + Delta
```

For a portfolio with weight vector w, the portfolio factor exposure is X'_p = w'X, and portfolio return attributes cleanly into a factor contribution and a specific contribution:

```
R_p = X'_p f + w'_p u
Var(R_p) = X'_p F X_p + w'_p Delta w_p
```

with the first term capturing factor (systematic) risk and the second capturing specific (diversifiable) risk. This decomposition is the reason factor models are indispensable for both risk attribution -- understanding what is actually driving a portfolio's risk and return -- and portfolio construction, since it lets an optimizer reason about a small number of factor exposures rather than thousands of individual stock covariances.

2. Factor Exposure Matrix

2.1 Estimation Universe (ESTU)

The regression that estimates factor returns each day is run over a curated "estimation universe," not the full investable universe. Construction typically layers several filters: a cumulative market-cap threshold per country (capturing, say, the top X% of market cap), a sector-level cumulative threshold to avoid excluding an entire thin sector dominated by small-caps, a minimum-price filter (smoothed to avoid names flickering in and out near the threshold), de-duplication of linked share classes and depositary receipts down to one primary listing per issuer, and manually maintained inclusion/exclusion lists to handle edge cases such as dual-listed companies that would otherwise be miscounted.

2.2 Outlier Handling

Raw fundamental and market descriptors can contain extreme outliers that would otherwise distort both exposures and factor returns. A robust standard deviation -- the median absolute deviation scaled by 1.4826 so that it converges to the ordinary standard deviation under normality -- is used to identify and trim the most egregious outliers first (since the ordinary standard deviation is itself distorted by the very outliers it should detect). A second, conventional-standard-deviation trim is then applied to catch remaining moderate outliers.

2.3 Standardization

Each raw descriptor is converted to a z-score within its country (subtracting the equal-weighted mean and dividing by the equal-weighted standard deviation), then rigidly shifted so that the cap-weighted market portfolio has an exposure of exactly zero to every style factor -- i.e., the market portfolio is "style neutral" by construction. Composite factors (built from more than one descriptor) are formed by first blending the standardized descriptors with fixed weights, then re-standardizing the resulting composite the same way.

2.4 Style Factor Definitions

The style factors I reviewed generally fall into these families (construction always follows the same pattern: raw descriptor -> outlier trim -> z-score -> cap-weight-zero standardization):

Factor family	Representative descriptors	Interpretation
Value	12-month forward earnings yield (weighted blend of current- and next-fiscal-year consensus EPS over price); book-to-price; sales-to-price; cash-flow-to-price	Cheapness relative to fundamentals
Size	Log of market capitalization	Small-cap vs large-cap premium
Momentum	Trailing risk-adjusted price momentum over a lookback window (excluding the most recent month to avoid short-term reversal contamination)	Continuation of recent relative performance
Market Beta / Residual Volatility	Time-series beta to the local market; residual (beta-adjusted) return volatility -- split into two factors rather than one, versus an earlier-generation single volatility factor	Systematic vs idiosyncratic volatility, separated
Liquidity	Share turnover (EWMA of traded volume / shares outstanding); bid-ask spread (sign-flipped so that tighter spreads load positively)	Ease of trading the name
Growth	Historical 5-year sales/earnings growth trend (regression slope, normalized by mean assets); forward analyst-estimated medium-term and long-term earnings growth	Fundamental growth trajectory
Variability / Quality	Standard deviation of annual net income, sales, and operating cash flow over 5 years, each normalized by mean total assets	Earnings/cash-flow stability (low variability approx. higher quality)
Profitability, Leverage, Dividend Yield, and others	Return on assets/equity; total debt to assets or equity; trailing and forward dividend yield; further descriptors follow the identical construction pattern	Additional well-documented return/risk drivers

2.5 Industry and Country Beta Factors

Rather than assigning every stock a fixed unit exposure to its industry (the traditional 0/1 dummy-variable approach), industry and country betas are estimated by regressing each stock's local excess return against the cap-weighted return of its industry or country grouping, using weighted least squares with more weight on recent observations. The choice of industry granularity is an empirical trade-off: too granular (hundreds of sub-industries) creates thin, noisy factor returns; too coarse (a handful of broad sectors) loses explanatory power. The betas are trimmed for outliers and rigidly shifted so the cap-weighted mean industry/country beta is exactly 1, preserving the intuitive reading that the market portfolio has unit exposure to each industry in proportion to its actual market weight.

A related concept is the *satellite country factor*: within a single-country model, a neighbouring market that is small relative to the home market (e.g., Canada inside a US model) can be represented as one additional factor rather than a full local factor set, capturing its distinct behaviour without fragmenting the core estimation universe.

2.6 Coverage Universe and Linked Securities

Local models use a detailed local factor set for one market; the global model spans all markets under one parsimonious factor set; the integrated model combines both. Linked securities -- different share classes or depositary receipts representing the same underlying issuer -- are collapsed to a single primary listing for the purpose of computing exposures, since including every linked line item would double-count the same fundamental risk.

3. Factor Returns

3.1 Cross-Sectional Regression

Factor returns are re-estimated every trading day by cross-sectional regression of that day's local excess stock returns against the (already known, previous-day) factor exposures:

```
r = X f + u (solved each day for f)
```

The regression uses weighted least squares, and the solution can be written explicitly as a matrix of "pure factor portfolio" weights Omega applied to the return vector, f = Omega r, where each row of Omega is the holdings vector of a portfolio with exactly unit exposure to one factor and zero exposure to every other factor. A pure factor portfolio's return is, by construction, precisely that factor's return for the day -- this is what makes factor returns directly interpretable as investable portfolio returns.

3.2 Regression Weights and Return Winsorization

As noted in Section 1.1, the regression is weighted by inverse residual variance rather than the traditional square-root-of-market-cap convention, since this is the weighting that minimizes the sampling noise of the factor-return estimate. Extreme individual stock returns are winsorized before the regression; the trimming thresholds are calibrated so that they meaningfully reduce the influence of extreme small-cap returns on the size factor's long-run drift without distorting the pure factor portfolios' overall behaviour.

3.3 Constraints and Interpretation of Pure Factor Portfolios

Because industry (and, in multi-country models, country) exposures are linear combinations of each other plus the market factor, the regression is under-identified without constraints. The standard fix is to require that cap-weighted industry factor returns sum to zero (and, in multi-country models, that cap-weighted country factor returns also sum to zero), which restores a unique solution and -- conveniently -- ensures that the market factor return is interpretable as (approximately) the cap-weighted market portfolio's own local excess return, since the specific-return contribution to the market portfolio is diversified away and typically negligible.

3.4 Satellite Factor Returns

For a satellite country factor (Section 2.5), the return is estimated in two steps: first, residualize the satellite country's local excess stock returns against the market, industry, and style factors already in the model; second, run a univariate cross-sectional regression of those residuals against the stock's country beta to isolate the satellite factor's own return.

4. Factor Volatilities

4.1 The Term Structure of Risk

The naive "square-root-of-time" rule -- scaling a one-day volatility (sigma) to a T-day volatility of sigma x sqrt(T) -- is only exactly correct when daily returns are independent and stationary. In reality, daily factor returns often exhibit serial correlation: momentum-type factors tend to show positive serial correlation (volatility increases faster than sqrt(T) as the horizon lengthens), while some factors (and many equity indices net of a strong mean-reverting component) can show negative serial correlation (volatility grows more slowly than sqrt(T)). A simple simulation makes this vivid: paths simulated with correlation of -0.5 stay far inside the +/-2-sigma-sqrt(T) envelope implied by the naive rule, while paths simulated with correlation of +0.5 breach it constantly.

Because different investors care about different horizons -- a risk manager with a short horizon, a portfolio manager thinking in quarters, and an asset owner making strategic allocation decisions over years -- a model built for one purpose is a poor fit for the others. This is the rationale for calibrating six distinct horizon-specific models (daily through long-term) sharing one factor structure but differing in how their covariance and specific-risk parameters are estimated: shorter-horizon models are calibrated primarily for forecasting accuracy at their own horizon (responsive, less stable); the annual model weights stability more heavily, subject to a floor on responsiveness; and the long-term model is deliberately built for maximum stability, using very long-memory (EWMA half-life) parameters.

4.2 Estimating the Term Structure: Two Complementary Methods

Low-frequency method. Daily factor returns are aggregated into rolling T-day windows (T matched to the horizon for the weekly and monthly models; shorter than the horizon for longer models, since most of the serial-correlation effect is captured by a shorter window while reducing sampling error relative to very long windows), and the variance of these aggregated returns is estimated with an exponentially weighted moving average (EWMA), so recent observations get more weight and the model can adapt to changing volatility regimes.

Newey-West method. A complementary technique explicitly estimates the covariance contribution of returns across different lagged trading days (a heteroskedasticity- and autocorrelation-consistent estimator), rather than relying purely on aggregation windows.

Blending the two. Each method has a different bias/noise trade-off, and a weighted blend of two independently biased-and-noisy estimates can have materially lower root-mean-square forecast error than either estimate alone. Algebraically, for two forecasts with bias mu_1, mu_2, noise eps_1, eps_2 and error correlation rho, the blended forecast var_B = w1 x var_hat_1 + (1-w1) x var_hat_2 has an RMS error that is minimized at some interior blending weight w1 between 0 and 1 whenever the two forecast errors are not perfectly positively correlated -- the benefit is largest when the errors are uncorrelated or negatively correlated, and more modest when they move together.

```
w1_optimal minimizes E[(w1 x var_hat_1 + (1-w1) x var_hat_2 - 1)^2] given
(mu_1,eps_1,mu_2,eps_2,rho)
```

4.3 Cross-Sectional Volatility (CSV) Scaling

There is a second, orthogonal tension in volatility forecasting: sampling error (favoring a long EWMA half-life) versus non-stationarity (favoring a short half-life, since the recent past is the best guide to the immediate future). The half-life parameter alone cannot fully resolve this trade-off, so a cross-sectional bias-correction technique is layered on top: at each point in time, the standardized cross-section of realized returns (z-scores) is used to detect "instantaneous" biases in the current risk forecasts, which then feed back into a multiplicative scaling adjustment on the volatility forecast. This materially reduces the tendency of EWMA-based models to under-forecast risk heading into a crisis and to stay elevated too long after a crisis subsides -- a stylized example: during an equity-market crisis, an unadjusted model's bias statistic (realized/predicted volatility) can peak noticeably higher than the CSV-adjusted version, and can likewise remain more depressed than the adjusted version once volatility has normalized.

This adjustment is not just a heuristic: it can be shown analytically that the cross-sectional scaling factor is exactly the multiplicative adjustment that minimizes the realized Q-statistic (see the Appendix) over a trailing window -- i.e., CSV scaling is the provably risk-minimizing correction given the available cross-sectional information.

5. Factor Correlations

5.1 Term Structure of Correlations

Just like volatilities, factor correlations exhibit their own term structure, and are estimated with the same two complementary techniques described in Section 4.2 -- a low-frequency method using rolling T-day aggregation windows, blended with a Newey-West estimator that explicitly accounts for cross-day lead/lag covariances.

5.2 Correlation Matrices Must Serve Two Different Purposes

A risk model's correlation matrix is used for two distinct purposes: (1) forecasting the volatility of a given (typically non-optimized) portfolio, and (2) as an input to portfolio optimization. A matrix that is a faithful, low-bias estimate of pairwise correlations is not automatically well-conditioned enough for optimization -- if the number of factors is large relative to the estimation history (exactly the multi-asset-class problem revisited from Section 1.3), the matrix can become ill-conditioned, with unrealistically small eigenvalues that an optimizer will aggressively (and spuriously) lever into oversized "safe" combinations.

One well-known alternative approach for multi-asset-class correlation estimation runs a pure time-series regression to explain local-factor correlations from a smaller set of global drivers. In practice this method has been shown to *systematically underestimate* cross-asset-class correlations -- sometimes very substantially (illustratively, a true correlation of 0.50 between an equity and a fixed-income factor being estimated as low as 0.15) -- because the global drivers cannot fully identify the core comovement across very different asset classes.

PCA shrinkage. A more robust alternative blends the sample (Newey-West-augmented) correlation matrix with a correlation matrix reconstructed from its own leading principal components -- i.e., using PCA to capture the small number of dominant common drivers of comovement, which by construction has better-behaved eigenvalues, and blending it back with the original sample matrix so that the result deviates only minimally from the best direct estimate:

```
C_tilde = w_B x C_B + (1-w_B) x C_PC
```

where C_B is the blended (low-frequency + Newey-West) sample correlation matrix and C_PC is its PCA-derived counterpart. In an illustrative empirical comparison, PCA shrinkage fit realized sample correlations far more tightly than the pure time-series method across a large panel of equity and fixed-income factor pairs (mean absolute error roughly 0.02 versus roughly 0.05 for the pure time-series approach).

5.3 Model Integration: Preserving Local Diagonal Blocks

When multiple local models are combined into an integrated multi-country or multi-asset-class model, each local model already has its own best-estimate correlation matrix (the "diagonal block"). Simply inserting the local matrices as diagonal blocks into the larger global matrix risks breaking positive semi-definiteness, since positive-definiteness is a joint property of the whole matrix, not something that can be guaranteed block by block. The practical solution is to compute a single global correlation matrix spanning all local models via the PCA-shrinkage approach, replace its diagonal blocks with the (already high-quality) local model matrices, and then apply the minimum necessary adjustment to the off-diagonal cross-model blocks to restore positive-definiteness -- preserving the trusted local estimates exactly while making the smallest possible change elsewhere.

6. Specific Risk Model

6.1 Time-Series Forecasts

Specific (idiosyncratic, non-factor) returns u_nt = r_nt - sum_k(X_nk x f_kt) are the residuals left over once factor returns are estimated. They matter most for concentrated or poorly diversified portfolios. Given a stock's history of specific returns, the same term-structure techniques used for factor volatility (Section 4.2) apply directly to forecast specific-risk term structure.

6.2 Structural Forecasts (for Short-History Names)

Not every stock has a full history of specific returns -- recent IPOs are the obvious example. For these, a structural model assigns a specific-risk forecast based on the stock's factor exposures: the log of specific volatility is regressed cross-sectionally against the factor exposure matrix,

```
ln(sigma_TS) = X b + epsilon
```

and the structural estimate is recovered by exponentiating the fitted (residual-free) part, with a small multiplicative bias correction c (needed because of Jensen's inequality: exponentiating a mean-zero residual does not, on average, give exactly 1):

```
sigma_STR = c x exp(X b)
```

In practice, structural forecasts track time-series forecasts quite closely for names that have both, which is a useful validation of the structural model's reasonableness.

6.3 Blended Forecasts, CSV Scaling, and Missing Data

As with factor volatility, blending the time-series and structural forecasts (weighted to minimize expected forecast error) generally produces a more accurate specific-risk estimate than either alone, and the same cross-sectional volatility (CSV) scaling technique from Section 4.3 is applied at the specific-risk level to further reduce bias using recent cross-sectional information.

6.4 Linked Specific Risk

The assumption that specific returns are mutually uncorrelated breaks down for linked securities -- different share classes or listings of the same underlying issuer. A no-arbitrage argument means these securities' returns must track each other closely (though not in perfect lock-step, since short-run liquidity differences and asynchronous trading hours can drive small, temporary wedges). The model therefore explicitly permits non-zero specific-return correlation between linked securities of the same issuer, rather than forcing the usual diagonal-only assumption.

7. The Finite-Sample Adjustment

7.1 The Problem: Double-Counting Specific Risk

The traditional asset covariance matrix Omega_hat = X F_hat X' + Delta_hat, built from the estimated factor covariance F_hat and the estimated diagonal specific-variance matrix Delta_hat, has a subtle but important flaw. A pure factor portfolio (unit exposure to its own factor, zero exposure to every other factor) already has an observed return series that reflects both the true (unobservable) factor return and its own true (unobservable) specific return -- the factor-return estimate f_hat is itself already "contaminated" by a share of specific risk, precisely because the estimation universe is finite (only N stocks, not infinitely many). F_hat = cov(f_hat) is therefore already an unbiased estimate of the pure factor portfolio's own variance, idiosyncratic component and all.

When Omega_hat = X F_hat X' + Delta_hat is then applied back to that same pure factor portfolio to "forecast" its risk, the model adds the specific-variance term w'_k Delta_hat w_k on top of a factor-variance term that already implicitly contains that specific-risk contribution -- double-counting it, and systematically overstating the risk of pure factor portfolios (and, by extension, any portfolio that is materially tilted toward a small number of factors, such as many optimized or concentrated portfolios).

```
w'_k Omega_hat w_k = F_hat_kk + w'_k Delta_hat w_k [the second term is spurious
double-counting]
```

A simple stylized example makes the size of the effect concrete: if the true factor variance is 1 and the true specific-return variance is epsilon, the naive approach can overshoot the true factor variance by epsilon/2 while simultaneously *under*-estimating the true specific variance by a factor of 2 -- the two errors partially offset in aggregate but distort the factor/specific risk split, which matters greatly for risk attribution and for any optimizer that treats the two risk sources differently.

7.2 The Consequences

- Systematic overstatement of risk for optimized or factor-tilted portfolios.
- Reduced realized out-of-sample risk-adjusted performance, since an optimizer working against an inflated risk estimate makes worse trade-offs.
- Distorted incentives around portfolio leverage and turnover, as the optimizer is implicitly fighting a phantom risk premium.

7.3 The Fix: Finite-Sample Adjustment (FSA)

The finite-sample adjustment re-derives unbiased estimates of the true factor covariance F and true specific-variance matrix Delta (rather than the naively double-counted versions), such that applying the resulting FSA asset covariance matrix Omega_tilde back onto the very pure factor portfolios that defined the model exactly reproduces their own already-unbiased observed variance -- by construction, no double-counting:

```
Omega_tilde = X F X' + Delta such that w'_k Omega_tilde w_k = F_hat_kk (no added Delta
term)
```

This is entirely a finite-sample effect: with an infinite estimation universe, a pure factor portfolio's specific risk would diversify away to zero and no adjustment would be needed at all -- the correction is driven purely by having a finite number of stocks in the estimation universe. In an illustrative daily-horizon comparison, the FSA-adjusted specific-risk forecasts for pure factor portfolios showed a materially tighter mean bias statistic (close to 1.0) than the unadjusted forecasts across a full sample period, confirming the correction works as intended in practice, not just in theory.

Appendix: Model Validation -- Bias and Q-Statistics

A.1 The Bias Statistic

For a factor (or stock) return f_k,t with beginning-of-period volatility forecast sigma_k,t, define the standardized return (z-score):

```
z_k,t = f_k,t / sigma_k,t
```

If the volatility forecast is unbiased and the z-scores are approximately mean-zero, the **bias statistic** -- the standard deviation of the z-scores over a trailing window -- should be close to 1.0. A bias statistic persistently above 1 means the model is under-forecasting risk; persistently below 1 means it is over-forecasting.

A.2 The Q-Statistic

A complementary and more information-rich validation metric is the Q-statistic, defined per period as Q = z^2 - ln(z^2), averaged across factors and time. Unlike the bias statistic, the Q-statistic is provably minimized in expectation exactly when the volatility forecast equals the true volatility (regardless of the underlying return distribution), which makes it attractive as an objective function for calibrating model parameters (it is, in fact, the quantity that CSV scaling and forecast-blending weights are chosen to minimize).

A distinctive and useful property of the Q-statistic is that it penalizes under-forecasting more heavily than over-forecasting of the same relative magnitude. In a stylized example, over-forecasting volatility by 10% increases the expected Q-statistic by about 0.017, while under-forecasting by the same 10% increases it by about 0.024 -- and the asymmetry grows sharply at larger errors (a 100% overshoot versus a 50% undershoot is not remotely symmetric in its Q-statistic impact). As a rule of thumb, a reduction of about 0.02 in the mean Q-statistic when comparing two candidate models is considered a materially better risk forecast; even a 0.01 reduction (roughly a 7% cut in forecast error) is arguably worth having.

Closing Note

These notes deliberately paraphrase and re-derive the general design principles of a modern multi-asset-class equity factor risk model in my own words, for my own study -- they are not a reproduction of any vendor's proprietary text. The accompanying **Kaushik_KAC3_Factor_Model_Example.xlsx** workbook applies the same core ideas -- cross-sectional regression, pure factor portfolios, and factor-covariance-based portfolio risk -- on a small illustrative 10-stock universe, with the understanding that a production-grade implementation additionally re-estimates factor exposures periodically (rather than holding them fixed across the sample) and layers on the term-structure, CSV-scaling, and finite-sample-adjustment refinements described above.